C200 Programming Assignment № 10

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Introduction

This is an **optional** homework. **If you choose to work on it, you must complete this on your own.** We're providing minimal discreption. Do not use any resources, human or computer.

Make sure to submit to Autograder and push your work by **December 11, 11:00 PM EDT, Wednesday**

1: Overlap

The function combine takes a list of intervals $[[i_0, i_1], [j_0, j_1], [k_0, k_1], \ldots]$ where either $i_1 < j_0$ or $i_1 >= j_0$. If the latter condition is true, then combine the two intervals: $[[i_0, j_1], [k_0, k_1], \ldots]$ and compare j_1 and k_0; otherwise keep $[[i_0, i_1]]$ and analyze $[j_0, j_1], [k_0, k_1], \ldots]$.

The following code:

```
lst = [[[1,3]],
       [[1,3],[4,5]],
       [[1,3],[2,6],[8,11],[9,12]],
       [[1,2],[2,3],[3,4],[4,12]],
       [[1,2],[3,4],[5,6]]]
for el in lst:
    print(f"{el} -> {combine(el)}")
```

produces:

```
[[1, 3]] -> [[1, 3]]
[[1, 3], [4, 5]] ->  [[1, 3], [4, 5]]
[[1, 3], [2, 6], [8, 11], [9, 12]] -> [[1, 6], [8, 12]]
[[1, 2], [2, 3], [3, 4], [4, 12]] -> [[1, 12]]
[[1, 2], [3, 4], [5, 6]] -> [[1, 2], [3, 4], [5, 6]]
```

Programming Problem 1: Combine

- Complete the function
- You are free to implement in any way, but my solution is recursive (which seems simpler)
- Assume there's at least one interval

Problem 2: Balance

Given a possibly empty list of numbers $[z_0, z_1, \ldots, z_n]$ return the smallest k such that:

$$\sum_{i=0}^{k-1} z_i \quad = \quad \sum_{i=k}^{n} z_i \tag{1}$$

If no number exists, then return -1. The following code:

```python
data = [[1,2,3],[0,1,1,0,1,1],[0],[3,3],[1,2,3,3,2,1],[1,2,3,3,2,6]]

for d in data:
    ptr = b(d)
    print(ptr, d[0:ptr],d[ptr:],sum(d[0:ptr])-sum(d[ptr:]))
```

produces:

```
2 [1, 2] [3] 0
4 [0, 1, 1, 0] [1, 1] 0
-1 [] [0] 0
1 [3] [3] 0
3 [1, 2, 3] [3, 2, 1] 0
-1 [1, 2, 3, 3, 2] [6] 5
```

Observe, for example, 1+2 = 3 and 1+2+3 = 3+2+1; however, the last list doesn't not have a sum.

Problem 3: Stars

Here are two recursive functions for $n = 0, 1, 2, \ldots$

$$\star(0) \quad = \quad 1 \tag{2}$$

$$\star(n) \quad = \quad n \times \star(n-1) \tag{3}$$

$$\star\star(0) \quad = \quad 1 \tag{4}$$

$$\star(n) \quad = \quad \star(n) \times \star\star(n-1) \tag{5}$$

```python
    for i in range(10):
        print(i, star_star(i))

    0 1
    1 1
    2 2
    3 12
```

```
4 288
5 34560
6 24883200
7 125411328000
8 5056584744960000
9 1834933472251084800000
```

Programming Problem 2: Balance

- Complete the function
- You are free to implement in any way